

# Homework 7.2

## Problem 4

Verify that the following formula is exact for polynomials of degree  $\leq 4$ .

$$\int_0^1 f(x)dx \approx \frac{1}{90} \left[ 7f(0) + 32f\left(\frac{1}{4}\right) + 12f\left(\frac{1}{2}\right) + 32f\left(\frac{3}{4}\right) + 7f(1) \right]$$

Let  $p(x) = ax^4 + bx^3 + cx^2 + dx + e$

$$\int_0^1 [ax^4 + bx^3 + cx^2 + dx + e]dx = \frac{a}{5} + \frac{b}{4} + \frac{c}{3} + \frac{d}{2} + e$$

$$p(0) = e \Rightarrow 7p(0) = 7e$$

$$p\left(\frac{1}{4}\right) = \frac{a}{256} + \frac{b}{64} + \frac{c}{16} + \frac{d}{4} + e \Rightarrow 32p\left(\frac{1}{4}\right) = \frac{a}{8} + \frac{b}{2} + 2c + 8d + 32e$$

$$p\left(\frac{1}{2}\right) = \frac{a}{16} + \frac{b}{8} + \frac{c}{4} + \frac{d}{2} + e \Rightarrow 12p\left(\frac{1}{2}\right) = \frac{3a}{4} + \frac{3b}{2} + 3c + 6d + 12e$$

$$p\left(\frac{3}{4}\right) = \frac{81a}{256} + \frac{27b}{64} + \frac{9c}{16} + \frac{3d}{4} + e \Rightarrow 32p\left(\frac{3}{4}\right) = \frac{81a}{8} + \frac{27b}{2} + 18c + 24d + 32e$$

$$p(1) = a + b + c + d + e \Rightarrow 7p(1) = 7a + 7b + 7c + 7d + 7e$$

$$\begin{aligned} & \left[ 7p(0) + 32p\left(\frac{1}{4}\right) + 12p\left(\frac{1}{2}\right) + 32p\left(\frac{3}{4}\right) + 7p(1) \right] \\ &= a \left[ \frac{7}{8} + \frac{6}{8} + \frac{81}{8} + \frac{56}{8} \right] + b \left[ \frac{7}{2} + \frac{3}{2} + \frac{27}{2} + \frac{14}{2} \right] + c[2c + 3c + 18c + 7c] \\ &+ d[8 + 6 + 24 + 7] + e[7 + 32 + 12 + 32 + 7] \end{aligned}$$

$$\left[ 7p(0) + 32p\left(\frac{1}{4}\right) + 12p\left(\frac{1}{2}\right) + 32p\left(\frac{3}{4}\right) + 7p(1) \right] = a[18] + b \left[ \frac{45}{2} \right] + c[30] + d[45] + e[90]$$

$$\Rightarrow \frac{1}{90} \left[ 7p(0) + 32p\left(\frac{1}{4}\right) + 12p\left(\frac{1}{2}\right) + 32p\left(\frac{3}{4}\right) + 7p(1) \right] = a \left[ \frac{18}{90} \right] + b \left[ \frac{\frac{45}{2}}{90} \right] + c \left[ \frac{30}{90} \right] + d \left[ \frac{45}{90} \right] + e \left[ \frac{90}{90} \right]$$

$$\therefore \int_0^1 p(x)dx \approx \frac{a}{5} + \frac{b}{4} + \frac{c}{3} + \frac{d}{2} + e$$

Since we have that the integral matches up with our terms in the approximation and since our coefficients are arbitrary, we have that this approximation is accurate for polynomials of degree  $\leq 4$ .

## Problem 6

Calculate  $\ln(2)$  approximately by applying the formula in the preceding problem to

$$\int_0^1 \frac{dt}{t+1}$$

Compare your answer to the correct value and compute the error.

$$f(t) = \frac{1}{t+1}$$

$$f(0) = 1 \Rightarrow 7f(0) = 7$$

$$f\left(\frac{1}{4}\right) = \frac{4}{5} \Rightarrow 32f\left(\frac{1}{4}\right) = \frac{128}{5}$$

$$f\left(\frac{1}{2}\right) = \frac{2}{3} \Rightarrow 12f\left(\frac{1}{2}\right) = 8$$

$$f\left(\frac{3}{4}\right) = \frac{4}{7} \Rightarrow 32f\left(\frac{3}{4}\right) = \frac{128}{7}$$

$$f(1) = \frac{1}{2} \Rightarrow 7f(1) = \frac{7}{2}$$

$$\int_0^1 \frac{dt}{t+1} = \ln(2) \approx \frac{1}{90} \left[ 7 + \frac{128}{5} + 8 + \frac{128}{7} + \frac{7}{2} \right] = \frac{4367}{6300} \approx .69317460$$

$$\ln(2) = .693147805599453094172321214581 \dots$$

$$\left| \ln(2) - \frac{4367}{6300} \right| = 2.7422614657865 \dots \times 10^{-5}$$

$$\frac{\left| \ln(2) - \frac{4367}{6300} \right|}{|\ln(2)|} = 3.956247017511 \dots \times 10^{-5}$$

## Problem 12

Derive a formula for approximating

$$\int_1^3 f(x) dx$$

In terms of  $f(0)$ ,  $f(2)$  and  $f(4)$ . It should be exact for all  $f$  in  $\Pi_2$ .

$$\int_1^3 f(x) dx \approx A_0 f(0) + A_1 f(2) + A_2 f(4)$$

$$\int_1^3 1 dx = 2 = A_0 + A_1 + A_2$$

$$\begin{aligned}\int_1^3 x dx &= 4 = 0A_1 + 2A_1 + 4A_2 \\ \int_1^3 x^2 dx &= \frac{26}{3} = 0A_1 + 4A_1 + 16A_2 \\ \Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 4 & 16 \end{bmatrix} \begin{bmatrix} A_0 \\ A_1 \\ A_2 \end{bmatrix} &= \begin{bmatrix} 2 \\ 4 \\ \frac{26}{3} \end{bmatrix}\end{aligned}$$

Using MATLAB (Note that all these used MATLAB):

$$\begin{aligned}\begin{bmatrix} A_0 \\ A_1 \\ A_2 \end{bmatrix} &= \begin{bmatrix} \frac{1}{12} \\ \frac{11}{6} \\ \frac{1}{12} \end{bmatrix} \\ \therefore \int_1^3 f(x) dx &\approx \frac{1}{12}f(0) + \frac{11}{6}f(2) + \frac{1}{12}f(4)\end{aligned}$$

### Problem 13

Determine values for  $A, B$ , and  $C$  that make the formula

$$\int_0^2 xf(x) dx \approx Af(0) + Bf(1) + Cf(2)$$

Exact for some polynomial of degree as high as possible. What is the maximal degree?

$$f(x) = 1 \Rightarrow \int_0^2 x dx = 2 = A + B + C$$

$$f(x) = x \Rightarrow \int_0^2 x^2 dx = \frac{8}{3} = 0A + B + 2C$$

$$f(x) = x^2 \Rightarrow \int_0^2 x^3 dx = 4 = 0A + 1B + 4C$$

Since we have that we now can create a matrix that is invertible, we have that this is the highest we can go now. We will check later to see if it holds for higher polynomials.

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} 2 \\ \frac{8}{3} \\ 4 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{4}{3} \\ \frac{2}{3} \end{bmatrix}$$

$$\therefore \int_0^2 xf(x)dx \approx \frac{4}{3}f(1) + \frac{2}{3}f(2)$$

$$f(x) = x^3 \Rightarrow \int_0^2 x^4 dx = \frac{32}{5}$$

$$\frac{4}{3}f(1) + \frac{2}{3}f(2) = \frac{4}{3} + \frac{2}{3} * 8 = \frac{20}{3} \neq \frac{32}{5}$$

### Problem 31

Determine the minimum number of subintervals needed to approximate

$$\int_1^2 (x + e^{-x^2}) dx$$

To an accuracy of at least  $\frac{1}{2} * 10^{-7}$ , using the trapezoid rule.

Since we know that the error for the composite trapezoid rule is given by:

$$-\frac{1}{12}(b-a)h^2 f''(\xi), \xi \in (a,b)$$

Where  $a = 1$ ,  $b = 2$ , and  $h$  is our subdivision length, we get that:

$$\frac{1}{2} * 10^{-7} = \left| -\frac{1}{12}(2-1)h^2 f''(\xi) \right| = \frac{h^2}{12} \left| \left[ \frac{d^2}{dx^2}(x + e^{-x^2}) \right]_{x=\xi} \right|$$

Using calculus to find the max of the function on  $[1,2]$ .

$$\left[ \frac{d^2}{dx^2}(x + e^{-x^2}) \right]_{x=\xi} = e^{-\xi^2}(4\xi^2 - 2) \leq e^{-\left(\frac{\sqrt{3}}{2}\right)^2} \left( 4 \left( \frac{\sqrt{3}}{2} \right)^2 - 2 \right)$$

$$\Rightarrow \frac{1}{2} * 10^{-7} = \frac{h^2}{12} \left[ \frac{4}{e^{\frac{3}{2}}} \right]$$

$$\Rightarrow h^2 = \frac{\frac{3}{2} * 10^{-7}}{e^{\frac{3}{2}}} \Rightarrow h = \sqrt{\frac{\frac{3}{2} * 10^{-7}}{e^{\frac{3}{2}}}}$$

$$h = \frac{b-a}{n} \Rightarrow n = \frac{1}{h} \approx 1220$$

# Homework 7.3

## Problem 7

Find a formula of the form

$$\int_0^1 xf(x)dx \approx \sum_{i=0}^n A_i f(x_i)$$

With  $n = 1$ , that is exact for all polynomials of degree 3.

$$\int_0^1 xf(x)dx \approx A_0 f(x_0) + A_1 f(x_1)$$

$$f(x) = 1 \Rightarrow \int_0^1 x dx = \frac{1}{2} = A_0 + A_1 \Rightarrow A_0 = A_1 - \frac{1}{2}$$

$$f(x) = x \Rightarrow \int_0^1 x^2 dx = \frac{1}{3} = A_0 x_0 + A_1 x_1 \Rightarrow \frac{1}{3} = A_1(x_0 + x_1) - \frac{1}{2}x_0 \Rightarrow A_1 = \frac{\frac{1}{3} - \frac{1}{2}x_0}{x_0 + x_1}$$

$$A_0 = \frac{\frac{1}{3} - \frac{1}{2}x_0}{x_0 + x_1} - \frac{1}{2} \Rightarrow \frac{\frac{1}{3} - \frac{1}{2}x_0 - \frac{1}{2}x_0 - \frac{1}{2}x_1}{x_0 + x_1} = \frac{\frac{1}{3} - x_0 - \frac{1}{2}x_1}{x_0 + x_1}$$

$$f(x) = x^2 \Rightarrow \int_0^1 x^3 dx = \frac{1}{4} = A_0 x_0^2 + A_1 x_1^2 \Rightarrow \frac{1}{4} = \left( \frac{\frac{1}{3} - x_0 - \frac{1}{2}x_1}{x_0 + x_1} \right) x_0^2 + \left( \frac{\frac{1}{3} - \frac{1}{2}x_0}{x_0 + x_1} \right) x_1^2$$

$$f(x) = x^3 \Rightarrow \int_0^1 x^4 dx = \frac{1}{5} = A_0 x_0^3 + A_1 x_1^3 \Rightarrow \frac{1}{5} = \left( \frac{\frac{1}{3} - x_0 - \frac{1}{2}x_1}{x_0 + x_1} \right) x_0^3 + \left( \frac{\frac{1}{3} - \frac{1}{2}x_0}{x_0 + x_1} \right) x_1^3$$

$$f_1(x_0, x_1) = \left( \frac{\frac{1}{3} - x_0 - \frac{1}{2}x_1}{x_0 + x_1} \right) x_0^2 + \left( \frac{\frac{1}{3} - \frac{1}{2}x_0}{x_0 + x_1} \right) x_1^2 = \frac{1}{4}$$

$$f_2(x_0, x_1) = \left( \frac{\frac{1}{3} - x_0 - \frac{1}{2}x_1}{x_0 + x_1} \right) x_0^3 + \left( \frac{\frac{1}{3} - \frac{1}{2}x_0}{x_0 + x_1} \right) x_1^3 = \frac{1}{5}$$

Using a solver to find  $x_0$  and  $x_1$ , we can then solve for  $A_0$  and  $A_1$ .

[Did not finish ...]

## Problem 11

For what values  $\alpha$  is this formula exact on  $\Pi_3$ ?

$$\int_0^2 f(x) dx \approx f(\alpha) + f(2 - \alpha)$$

$$f(x) = 1 \Rightarrow \int_0^2 1 dx = 2 = 1 + 1$$

$$f(x) = x \Rightarrow \int_0^2 x dx = 2 = \alpha + 2 - \alpha$$

$$f(x) = x^2 \Rightarrow \int_0^2 x^2 dx = \frac{8}{3} = \alpha^2 + 4 - 4\alpha + \alpha^2 = 2\alpha^2 - 4\alpha + 4 \Rightarrow \alpha^2 - \alpha + 2 - \frac{4}{3} = 0$$

$$f(x) = x^3 \Rightarrow \int_0^2 x^3 dx = 4 = \alpha^3 + (8 - 8\alpha + 2\alpha^2 - 4\alpha + 4\alpha^2 - \alpha^3) = 6\alpha^2 - 12\alpha + 4 = 0$$

$$\therefore \alpha = 1 \pm \frac{1}{\sqrt{3}}$$

## Problem 16

Using only  $f(0)$ ,  $f'(-1)$ , and  $f''(1)$ , compute an approximation of  $\int_{-1}^1 f(x) dx$  that is exact for all quadratic polynomials. Is the approximation exact for polynomials of degree 3? Why or why not?

$$f(x) = 1 \Rightarrow \int_{-1}^1 1 dx = 2 = A_0 1 + A_1 0 + A_2 0$$

$$f(x) = x \Rightarrow \int_{-1}^1 x dx = 0 = A_0 0 + A_1 1 + A_2 0$$

$$f(x) = x^2 \Rightarrow \int_{-1}^1 x^2 dx = \frac{2}{3} = A_0 0 + A_1 (-2) + A_2 2$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 2 \end{bmatrix} \begin{bmatrix} A_0 \\ A_1 \\ A_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ \frac{2}{3} \end{bmatrix} \Rightarrow \begin{bmatrix} A_0 \\ A_1 \\ A_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ \frac{1}{3} \end{bmatrix}$$

$$\therefore \int_{-1}^1 f(x) dx \approx 2f(0) + \frac{f''(1)}{3}$$

To see if this works for cubic polynomials:

$$f(x) = x^3 \Rightarrow \int_{-1}^1 x^3 dx = 0$$

$$2f(0) + \frac{f''(1)}{3} = 0 + 2 \neq 0$$